Decontamination Report for the First Cycle of OpenEuroLLM Pretraining Datasets

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Abstract

This report describes the decontamination process carried out for the first cycle of OpenEuroLLM pretraining datasets. Based on the NeMo Curator pipeline, covering 36 languages and 67 benchmarks, more than 11 billion documents are processed to identify n-gram overlaps between the benchmarks and the datasets. The results are analysed and used to assess the impact on model performance before and after the removal of contaminated documents, which is observed to be minimal.

1 Decontamination

We apply decontamination to the datasets that are part of the initial OpenEuroLLM pretraining data mix in order to mitigate leakage from downstream evaluation benchmarks into the training corpora. This step is critical given that LLMs are typically evaluated on benchmark datasets that are assumed to be unseen during training. If such benchmarks are present in the training set, evaluation results may be artificially inflated.

Our approach follows established practices used in previous large-scale models, where segments of training documents that overlap with benchmark data are identified and the documents containing them removed. We rely on an n-gram–based matching strategy to detect overlaps between the training corpus and a predefined set of benchmarks represeting downstream tasks.

1.1 Benchmarks for decontamination

The benchmark evaluation suite, shown in Table 1, spans multiple categories, including common sense reasoning, instruction following, mathematical reasoning, code generation, multilingual understanding, linguistic competence and world knowledge.

Common sense reasoning includes standard question answering (QA) and inference tasks such as COPA, HellaSwag, PIQA and related multilingual variants. Instruction following benchmarks such as AlpacaEval, MT-Bench or Toxigen evaluate general and safety-oriented adherence across diverse prompt-based benchmarks. Mathematical and coding abilities are measured using established datasets including GSM8K, MATH, MBPP or HumanEval. Additional categories assess multilingual and translation performance (FLORES-200), linguistic competence (MultiBlimp) and world knowledge (MMLU, GPQA, Belebele).

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Task	Category	Languages	Passages	Size (MB)
HumanEval	Code	1	328	10.6419
MBPP	Code	1	1352	4.8529
ARC Challenge	Common sense reasoning	1	3516	1.1415
ArcEasy	Common sense reasoning	1	7128	1.9972
BoolQ	Common sense reasoning	1	3270	2.0204
COPA	Common sense reasoning	1	1100	0.1422
CommonsenseQA	Common sense reasoning	1	1221	0.2635
HellaSwag	Common sense reasoning	1	10042	10.6699
Include	Common sense reasoning	44	8966	4.2247
Lambada	Common sense reasoning	1	5153	1.6316
MMMLU	Common sense reasoning	14	70210	38.149
OpenBookQA	Common sense reasoning	1	2000	0.3851
PIQA	Common sense reasoning	1	5512	1.4174
XCOPA	Common sense reasoning	11	2000	0.2371
XStoryCloze	Common sense reasoning	10	7555	2.5407
XWinograd	Common sense reasoning	6	2671	0.3142
AlpacaEval	Instruction	1	1593	0.1720
IFEval	Instruction	1	541	0.2743
MT-Bench-X	Instruction	5	400	0.2720
MTBench	Instruction	1	1388	1.8000
Multi-IFEval	Instruction	8	22505	34.3774
advbench	Instruction	1	500	0.3810
do_not_answer	Instruction	2	939	3.8161
m-ArenaHard	Instruction	23	6000	3.3170
Toxigen	Instruction	1	940	0.3500
PAWS-X	Language understanding	7	14000	3.4758
MultiBlimp	Linguistic	101	60693	92.1014
AIME 25	Math	1	30	0.0118
GSM8K	Math	1	1319	0.6815
Hendrycks_MATH	Math	1	5003	2.0000
MGSM	Math	7	1500	0.4786
PolyMath	Math	18	3000	0.6709
BasqueBench	Mix	1	43201	23.3597
CatalanBench	Mix	1	20259	10.6804
GalicianBench	Mix	1	10904	28.2443
SpanishBench	Mix	1	8964	24.7347
XNLI	Reasoning	15	27449	5.2555
FLORES-200	Translation	200	37444	10.3435
Belebele	World Knowledge	122	29700	27.5175
GPQA	World Knowledge	1	546	4.0179
Global MMLU	World Knowledge	23	210630	142.846
MMLU	World Knowledge	1	14042	6.6514

Table 1: Benchmark evaluation suite grouped by capability category, showing the amount of languages encompassed in each one, an approximate count of passages used for decontamination and the approximate size of the passages used.

The benchmark suite contains 67 individual benchmarks that comprise a single or multiple languages.

- **Multilingual**: ArcChallenge, ArcEasy, Belebele, COPA, Cocoteros, Flores200, GlobalMMLU_Full, Include, MGSM, MMMLU, MTBench_X, MultiBlimp, Multi_IFEval, OpenBookQA, PAWS, PAWS_x, PIQA, PolyMath, WNLI, XCOPA, XNLI, XStoryCloze, XWinograd, XWinogradNoBlanks, m_ArenaHard
- **English**: AIME_II_25, AIME_I_25, AlpacaEval, BoolQ, CommonsenseQA, DoNotAnswer, GPQA, GSM8K, HellaSwag, Hendrycks_MATH, HumanEval, HumanEval_Plus, IFEval, Lambada, MBPP, MBPP_Plus, MMLU, MTBench, ToxiGen

- **Basque**: EusExams, EusProficiency, EusReading, EusTrivia, QNLI
- **Catalan**: Cabreu, CatalanQA, Catcola, CoqCat, Parafraseja, SIQA, TECA, XNLIVA, XQUAD, PhrasesCAVA
- **Galician**: Galcola, Parafrases, Summarization, Truthfulqa
- **Spanish**: ESCOLA, Phrases, XQUAD, XLSUM

1.2 Decontamination methodology

We use the decontamination pipeline implemented in the NeMo Curator framework.[1] This method constructs n-gram representations of benchmark datasets and systematically removes any matching content from the training corpus to prevent contamination in three steps: indexing, matching and removal. The implementation is described in detail in [1] and [2], we summarize the main steps in the following section.

1.2.1 Indexing

The pipeline first normalizes and tokenizes the benchmark data, after which n-grams of size 8 to 13 are extracted to form a compact representation of the evaluation data. These n-grams are stored in an index and reused across all datasets that are being decontaminated.

The selection of source data for n-gram construction follows task-specific configurations defined by the lm-evaluation-harness framework [cite]. This ensures that the resulting benchmark n-gram index only reflects data intended for evaluation. In most cases, the *test* split is used, however, when one is not available, the corresponding *validation* split is used instead.

1.2.2 Matching

Contamination matching is performed for each dataset under consideration. Each document in the corpus is processed using the same pipeline applied during benchmark indexing, including normalization, tokenization, and extraction of n-grams of size 8 to 13.

The resulting n-grams are then compared against the precomputed task n-gram index to identify overlaps between the corpus and downstream evaluation data. For each matched n-gram, its frequency of occurrence within the corpus is recorded, yielding the statistics required for the subsequent filtering stage.

1.2.3 Removal

The final removal stage applies a frequency-based filtering criterion to the matched n-grams. N-grams with occurrence counts below a predefined threshold, chosen on a per-language basis, are treated as contamination signals and any documents containing them are completely removed from the training dataset.

1.3 Experiments

Decontamination procedures were conducted on a collection of datasets curated within the OpenEuroLLM training data catalogue framework. The resources used include FinePDFs and FinePDFs-edu, HPLT 3, Nemotron-CC (medium-high to high quality subsets), DCLM, and the OLMo-mix subsets: ArXiv, PeS2o, and Wikipedia, as well as the synthetically generated MultiSynt corpus. Together, these datasets provide a broad coverage of domains, formats, and quality levels suitable for large-scale language model training.

This collection of 7 corpora covers 36 languages in total. This includes the 24 official languages of the European Union, along with 3 co-official languages recognized within member states. It is further extended by 7 languages from EU candidate countries and 2 closely related Scandinavian languages, broadening the linguistic scope beyond the EU while maintaining regional relevance.

[1] https://docs.nvidia.com/nemo-framework/user-guide/25.07/datacuration/taskdecontamination.html

Dataset	Coverage	Documents
DCLM	English	2.939.202.307
Nemotron-CC	English	3.589.116.797
MultiSynt*	English	154.789.483
OLMo-mix	English	116.557.342
Finemath	English	6,699,493
Megamath-text-code-block	English	52,686,522
Megamath-web-pro	English	14,978,856
Starcoder	English	206,642,239
FinePDFs	Multilingual	384.703.437
FinePDFs-edu	Multilingual	41.934.960
HPLT v3	Multilingual	4.260.963.436

Table 2: Total amount of documents processed for all languages per corpus (HPLT v3 except English). *Decontaminated through Nemotron-CC.

Dataset	Documents	Contaminated	Contamination %
DCLM	2.939.202.307	540,043	0.0184 %
Nemotron-CC	3.589.116.797	635,095	0.0177 %
MultiSynt*	154.789.483	34,969	0.0002 %
OLMo-mix	116.557.342	19,060	0.0164 %
Finemath	6,699,493	18,703	0.2792 %
Megamath-text-code-block	52,686,522	9,927	0.0188 %
Megamath-web-pro	14,978,856	17,222	0.1150 %
Starcoder	206,642,239	5,769	0.0028 %
FinePDFs	384.703.437	101,860	0.0003 %
FinePDFs-edu	41.934.960	41,700	0.0010 %
HPLT v3	4.260.963.436	373,622	0.0001 %

Table 3: Per corpus contamination summary showing total amount of contaminated documents where at least one benchmark n-gram did not pass the threshold, and the contamination ratio of the corpus. *Decontaminated through Nemotron-CC.

Experiments were carried out on LUMI HPC, using NemoCurator's distributed worker pipeline, on *small*-type nodes with 512 GB RAM and 128 CPUs. The average decontamination time on these nodes for a $\approx$ 1GB *.jsonl* file is approximately 12 hours for Matching and Removal tasks together.

In total, we process 11.332.478.279 documents, as summarized in Table 2. As an outcome of the decontamination procedure, we identify the document-level IDs associated with contamination for each dataset. These identified documents are subsequently removed from the corresponding training sets, reducing the likelihood of overlap with downstream evaluation benchmarks, although some residual contamination may still remain due to the approximate nature of the tool used.

It must be noted that the Matching and Removal steps are skipped for all MultiSynt sets because the corpus is comprised of machine translated documents from Nemotron-CC. Given that document IDs are maintained for MultiSynt, simple post-filtering is sufficient to decontaminate the corpus using the contaminated IDs from Nemotron-CC.

1.4 Decontamination Analysis

The NeMo Curator decontamination pipeline operates largely as a black box and does not expose detailed intermediate metrics that would facilitate direct analysis of contamination patterns. To address this limitation, we persist the outputs of the decontamination process in JSONL format, where each line corresponds to a decontaminated example and includes the document identifier ("id") along with the full document text ("text"). This design enables posterior analysis, allowing us to approximate which benchmark-derived n-gram triggered the contamination flag for a given document, as well as to infer the originating benchmark dataset and text, with examples shown in Listing 1. Additionally, this stored output supports aggregate analyses, such as estimating the distribution of contaminated

documents across languages and benchmarks, and computing approximate contamination ratios by comparing flagged documents against the total number of documents within each dataset.

As shown in Table 3, the proportion of contaminated documents across the datasets considered remains consistently low. For DCLM, 540,043 documents are identified as contaminated, corresponding to 0.0184% of the dataset, while Nemotron-CC exhibits a similar level with 635,095 documents (0.0177%), as well as OLMo-mix subsets with 19,060 documents (0.0164%). FinePDFs and FinePDFs-edu show substantially lower contamination rates, with 101,860 (0.0003%) and 41,700 (0.0010%) documents respectively. HPLT v3 presents the lowest relative contamination, with 373,622 documents accounting for just 0.0001% of the dataset. Finally, for MultySynt 34,969 documents are identified and removed for each of the dataset's languages via postprocessing filtering using the same id's from Nemotron-CC English contaminated documents. Overall, these figures indicate that contamination is present but constitutes only a very small fraction of the total data.

Listing 1: Examples of matched contamination records for DCLM.


```
{"warc_record_id": "<urn:uuid:61d4359c-8698-4f07-8528-de7b79e55c2e>",
 "file_part": "local-shard_8_of_10_shard_00000235_processed",
 "benchmark": "BoolQ",
 "matched_ngram": "associated with aging and exposure to ultraviolet radiation from the sun they
      range",
 "benchmark_text": "Liver spot -- Liver spots (also known as age spot, solar lentigo, ''lentigo
      senilis'', ''old age spot'', ''senile freckle'') are blemishes on the skin associated with aging
      and exposure to ultraviolet radiation from the sun. They range in color from light brown to
      red or black and are located in areas most often exposed to the sun, particularly the hands,
      face, shoulders, arms and forehead, and the scalp if bald."}

{"warc_record_id": "<urn:uuid:9278fb1a-1ff4-4a4f-84cc-dbd5e885579d>",
 "file_part": "local-shard_6_of_10_shard_00000202_processed"
 "benchmark": "Belebele",
 "matched_ngram": "naval blockade of germany the strategy proved effective cutting off vital
      military and",
 "benchmark_text": "Soon after the outbreak of hostilities, Britain initiated a naval blockade of
      Germany. The strategy proved effective, cutting off vital military and civilian supplies,
      although this blockade violated generally accepted international law codified by several
      international agreements of the past two centuries. Britain mined international waters to
      prevent any ships from entering entire sections of ocean, causing danger to even neutral
      ships. Since there was limited response to this tactic, Germany expected a similar response
      to its unrestricted submarine warfare."}

 {"warc_record_id": "<urn:uuid:e3ba3273-f360-40ae-af8e-a80857d24a53>",
 "file_part": "local-shard_7_of_10_shard_00000224_processed",
 "benchmark": "MultiBlimp",
 "matched_ngram": "problem with this argument is that bush lacked the experience necessary to
      be",
 "benchmark_text": "The problem with this argument is that Bush lacked the experience necessary
      to be president when he ran in 2000, so this sort of cheap shot just hoists him by his own
      petard."}
```


Tables in Appendix A.1 show that contamination is unevenly distributed across benchmarks, although overall rates remain consistently low. For large web-scale corpora such as DCLM and Nemotron-CC, the highest concentrations of contamination are associated with widely used natural language understanding benchmarks, most notably BoolQ, which accounts for the largest share of flagged documents in both datasets. Additional contributions come from benchmarks such as HellaSwag, MMLU, and MTBench, although at substantially lower relative proportions. Despite the large absolute number of matches in some cases, the corresponding percentages remain insignificantly low, indicating that contamination is sparse relative to dataset size.

In contrast, curated and domain-specific corpora such as FinePDFs, FinePDFs-edu, and HPLT v3 exhibit markedly lower contamination ratios overall. Within these datasets, contamination is more strongly associated with multilingual and region-specific benchmarks (e.g., SpanishBench,

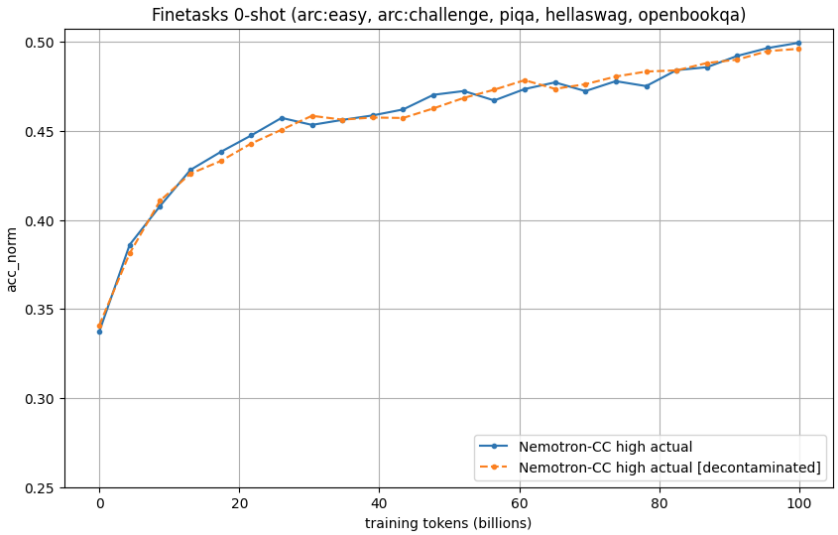


Figure 1: Performance comparison on Nemotron-CC with and without decontamination.

CatalanBench, and GalicianBench), reflecting the composition of the underlying data. Even in these cases, however, the relative proportions remain extremely small.

These results suggest that while contamination can be detected across a wide range of benchmarks and datasets, it is generally limited in scope and concentrated in specific evaluation sources rather than being broadly distributed.

1.5 Decontamination Experiments

To assess the impact of contamination on model performance, we conduct controlled training experiments using datasets before and after the decontamination process, keeping all training settings identical. In particular, we compare 0.6B-parameter models trained on Nemotron-CC and DCLM, and their decontaminated counterpart.

The training corpus is comprised approximately of 100B tokens, around 30M documents, where roughly 30K documents are removed during decontamination. The resulting models are evaluated using a consistent evaluation setup (oellm-cli) on dclm-core-22, covering a range of benchmarks including HellaSwag (both 0-shot and 10-shot), ARC-Easy, ARC-Challenge, OpenBookQA, and PIQA. This setup enables a direct comparison of performance and provides an empirical estimate of how the removal of contaminated data influences downstream evaluation results.

Across both Nemotron-CC and DCLM, the overall trend shows that decontamination has a minimal impact on aggregate downstream performance. The learning curves for contaminated and decontaminated variants track each other very closely throughout training, with differences typically within a narrow margin and no consistent advantage for either setting.

For Nemotron-CC, as seen in Figure 1, the two curves are almost indistinguishable across the full token range. At earlier stages (below $\approx$30B tokens), the differences fluctuate slightly but remain small. At later stages, the decontaminated model occasionally shows marginal improvements, though these are inconsistent and not clearly systematic. This suggests that the removal of 30K documents out of 30M has little measurable effect on model performance, even on benchmarks potentially affected by contamination.

A similar pattern is observed for DCLM, as seen in Figure 2. While there are minor deviations between the contaminated and decontaminated variants, particularly in the mid-training, these differences do not persist or compound as training progresses. By the end of training, both models converge to nearly identical performance levels.

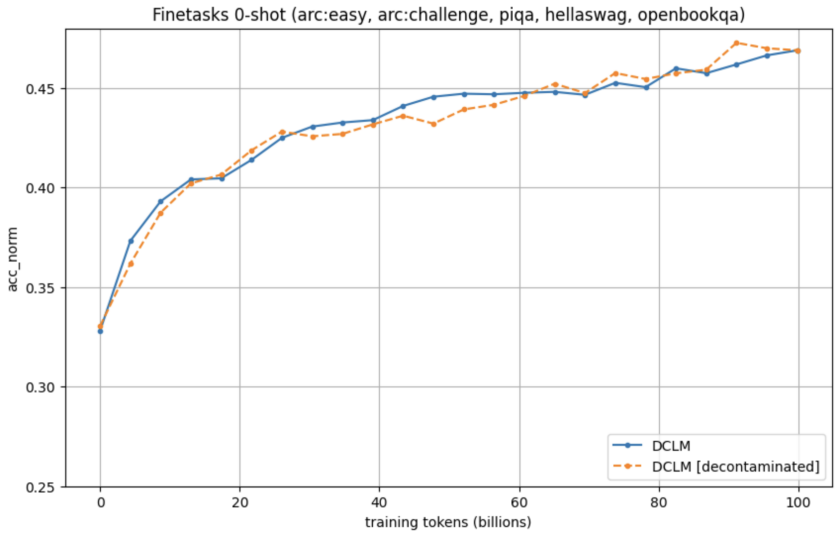


Figure 2: Performance comparison on DCLM with and without decontamination.

1.6 Contamination conclusions

The decontamination process relies on a systematic n-gram–based pipeline that detects and removes documents containing exact matches with benchmark-derived n-grams when identifying potential overlap. However, this approach does not capture paraphrased or semantically similar content. Within this framework, the overall fraction of removed documents remains small relative to the full corpus. Taken together, the resulting comparisons suggest that, at this scale and level of contamination, the benchmark overlap identified by this procedure has little observable effect on aggregate evaluation metrics, with any impact appearing minor relative to the inherent variability in training dynamics.

References

[1] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. *CoRR*, abs/2005.14165, 2020.

[2] Shaden Smith, Mostofa Patwary, Brandon Norick, Patrick LeGresley, Samyam Rajbhandari, Jared Casper, Zhun Liu, Shrimai Prabhumoye, George Zerveas, Vijay Korthikanti, Elton Zheng, Rewon Child, Reza Yazdani Aminabadi, Julie Bernauer, Xia Song, Mohammad Shoeybi, Yuxiong He, Michael Houston, Saurabh Tiwary, and Bryan Catanzaro. Using deepspeed and megatron to train megatron-turing NLG 530b, A large-scale generative language model. *CoRR*, abs/2201.11990, 2022.

A Technical appendices and supplementary material

A.1 Detailed contamination results

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	2	0.0000 %
AIME_I_25	0	0.0000 %
AlpacaEval	3798	0.0001 %
ArcChallenge	704	0.0000 %

Benchmark	Contaminated Matches	Contamination %
ArcEasy	1430	0.0000 %
Belebele	20697	0.0007 %
BoolQ	316594	0.0108 %
COPA	0	0.0000 %
CommonsenseQA	12	0.0000 %
DoNotAnswer	11	0.0000 %
Flores200	1545	0.0001 %
GPQA	2	0.0000 %
GSM8K	95	0.0000 %
HellaSwag	45970	0.0016 %
Hendrycks_MATH	5896	0.0002 %
HumanEval	165	0.0000 %
HumanEval_Plus	15	0.0000 %
IFEval	1479	0.0001 %
Lambada	3765	0.0001 %
m_ArenaHard	354	0.0000 %
MBPP	134	0.0000 %
MBPP_Plus	15	0.0000 %
MMLU	82725	0.0028 %
MTBench	39633	0.0013 %
MTBench_X	588	0.0000 %
MultiBlimp	5195	0.0002 %
Multi_IFEval	0	0.0000 %
OpenBookQA	1	0.0000 %
PAWS_x	3695	0.0001 %
PIQA	54	0.0000 %
PolyMath	1192	0.0000 %
ToxiGen	1137	0.0000 %
XNLI	2158	0.0001 %
XStoryCloze	8	0.0000 %
XWinograd	974	0.0000 %
XWinogradNoBlanks	0	0.0000 %

Table 4: Per benchmark contamination summary for DCLM dataset. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	7	0.0000 %
AIME_I_25	0	0.0000 %
AlpacaEval	3009	0.0001 %
ArcChallenge	793	0.0000 %
ArcEasy	1420	0.0000 %
Belebele	9313	0.0003 %
BoolQ	441547	0.0123 %
COPA	3	0.0000 %
CommonsenseQA	36	0.0000 %
DoNotAnswer	33	0.0000 %
Flores200	627	0.0000 %
GPQA	35	0.0000 %
GSM8K	181	0.0000 %
HellaSwag	50826	0.0014 %
Hendrycks_MATH	7260	0.0002 %
HumanEval	226	0.0000 %
HumanEval_Plus	2	0.0000 %
IFEval	1199	0.0000 %
Lambada	3172	0.0001 %
m_ArenaHard	402	0.0000 %
MBPP	394	0.0000 %
MBPP_Plus	105	0.0000 %
MMLU	58587	0.0016 %

Benchmark	Contaminated Matches	Contamination %
MTBench	46295	0.0013 %
MTBench_X	440	0.0000 %
MultiBlimp	2601	0.0001 %
Multi_IFEval	5	0.0000 %
OpenBookQA	10	0.0000 %
PAWS_x	3496	0.0001 %
PIQA	60	0.0000 %
PolyMath	580	0.0000 %
ToxiGen	521	0.0000 %
XNLI	821	0.0000 %
XStoryCloze	9	0.0000 %
XWinograd	1076	0.0000 %
XWinogradNoBlanks	4	0.0000 %

Table 5: Per benchmark contamination summary for Nemotron-CC dataset. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	3	0.000000 %
AIME_I_25	0	0.000000 %
AlpacaEval	543	0.000001 %
ArcChallenge	276	0.000001 %
ArcEasy	442	0.000001 %
BasqueBench	357	0.000001 %
Belebele	527	0.000001 %
BoolQ	20276	0.000053 %
COPA	4	0.000000 %
CatalanBench	876	0.000002 %
CommonsenseQA	80	0.000000 %
DoNotAnswer	10	0.000000 %
Flores200	30	0.000000 %
GPQA	47	0.000000 %
GSM8K	280	0.000001 %
GalicianBench	3461	0.000009 %
GlobalMMLU_Full	1750	0.000005 %
HellaSwag	706	0.000002 %
Hendrycks_MATH	2644	0.000007 %
HumanEval	115	0.000000 %
HumanEval_Plus	1	0.000000 %
IFEval	101	0.000000 %
Include	9933	0.000026 %
Lambada	284	0.000001 %
m_ArenaHard	50	0.000000 %
MBPP	43	0.000000 %
MBPP_Plus	10	0.000000 %
MGSM	1	0.000000 %
MMLU	18550	0.000048 %
MMMLU	2343	0.000006 %
MTBench	8824	0.000023 %
MTBench_X	190	0.000000 %
MultiBlimp	5950	0.000015 %
Multi_IFEval	40	0.000000 %
OpenBookQA	26	0.000000 %
PAWS_x	98	0.000000 %
PIQA	13	0.000000 %
PolyMath	528	0.000001 %
SpanishBench	21727	0.000056 %
ToxiGen	340	0.000001 %
XCopa	0	0.000000 %
XNLI	240	0.000001 %

Benchmark	Contaminated Matches	Contamination %
XStoryCloze	13	0.000000 %
XWinograd	124	0.000000 %
XWinogradNoBlanks	4	0.000000 %

Table 6: Per benchmark contamination summary for FinePDFs dataset for all languages. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	3	0.000000 %
AIME_I_25	0	0.000000 %
AlpacaEval	148	0.000004 %
ArcChallenge	237	0.000006 %
ArcEasy	340	0.000008 %
BasqueBench	27	0.000001 %
Belebele	224	0.000005 %
BoolQ	6995	0.000167 %
COPA	2	0.000000 %
CatalanBench	165	0.000004 %
CommonsenseQA	27	0.000001 %
DoNotAnswer	1	0.000000 %
Flores200	46	0.000001 %
GPQA	11	0.000000 %
GSM8K	166	0.000004 %
GalicianBench	523	0.000012 %
GlobalMMLU_Full	621	0.000015 %
HellaSwag	316	0.000008 %
Hendrycks_MATH	2179	0.000052 %
HumanEval	227	0.000005 %
HumanEval_Plus	1	0.000000 %
IFEval	32	0.000001 %
Include	3071	0.000073 %
Lambada	33	0.000001 %
m_ArenaHard	19	0.000000 %
MBPP	23	0.000001 %
MBPP_Plus	4	0.000000 %
MGSM	1	0.000000 %
MMLU	9499	0.000227 %
MMMLU	592	0.000014 %
MTBench	4485	0.000107 %
MTBench_X	149	0.000004 %
MultiBlimp	2998	0.000071 %
Multi_IFEval	7	0.000000 %
OpenBookQA	15	0.000000 %
PAWS_x	44	0.000001 %
PIQA	8	0.000000 %
PolyMath	315	0.000008 %
SpanishBench	7877	0.000188 %
ToxiGen	178	0.000004 %
XCopa	0	0.000000 %
XNLI	32	0.000001 %
XStoryCloze	3	0.000000 %
XWinograd	56	0.000001 %
XWinogradNoBlanks	0	0.000000 %

Table 7: Per benchmark contamination summary for FinePDFs-edu dataset for all languages. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
BasqueBench	484	0.000000 %
Belebele	516	0.000000 %
CatalanBench	15447	0.000004 %
Flores200	19	0.000000 %
GalicianBench	37185	0.000009 %
GlobalMMLU_Full	6865	0.000002 %
Include	17567	0.000004 %
m_ArenaHard	26	0.000000 %
MGSM	0	0.000000 %
MMMLU	7615	0.000002 %
MTBench_X	94	0.000000 %
MultiBlimp	17904	0.000004 %
Multi_IFEval	47	0.000000 %
PAWS_x	440	0.000000 %
PolyMath	8	0.000000 %
SpanishBench	269317	0.000063 %
XCopa	0	0.000000 %
XNLI	88	0.000000 %
XStoryCloze	0	0.000000 %
XWinograd	0	0.000000 %
XWinogradNoBlanks	0	0.000000 %

Table 8: Per benchmark contamination summary for HPLT v3 dataset for all languages. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	2	0
AIME_I_25	0	0
AlpacaEval	50	0
ArcChallenge	16	0
ArcEasy	25	0
Belebele	17	0
BoolQ	1510	0.0013
COPA	0	0
CommonsenseQA	16	0
DoNotAnswer	0	0
Flores200	6	0
GPQA	4	0
GSM8K	9	0
HellaSwag	11	0
Hendrycks_MATH	1195	0.001
HumanEval	93	0.0001
HumanEval_Plus	3	0
IFEval	23	0
Lambada	2	0
m_ArenaHard	33	0
MBPP	40	0
MBPP_Plus	10	0
MMLU	389	0.0003
MTBench	650	0.0006
MTBench_X	17	0
MultiBlimp	8	0
Multi_IFEval	0	0
OpenBookQA	2	0
PAWS_x	26	0
PIQA	8	0
PolyMath	203	0.0002
ToxiGen	1	0
XNLI	3	0
XStoryCloze	2	0

Benchmark	Contaminated Matches	Contamination %
XWinograd	3	0
XWinogradNoBlanks	0	0

Table 9: Per benchmark contamination summary for the arxiv subset of OLMo-mix dataset for English. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	0	0.0
AIME_I_25	0	0.0
AlpacaEval	55	0.0
ArcChallenge	10	0.0
ArcEasy	44	0.0
Belebele	19	0.0
BoolQ	1365	0.0012
COPA	0	0.0
CommonsenseQA	58	0.0
DoNotAnswer	0	0.0
Flores200	0	0.0
GPQA	8	0.0
GSM8K	20	0.0
HellaSwag	46	0.0
Hendrycks_MATH	69	0.0001
HumanEval	73	0.0001
HumanEval_Plus	0	0.0
IFEval	26	0.0
Lambada	12	0.0
m_ArenaHard	2	0.0
MBPP	8	0.0
MBPP_Plus	2	0.0
MMLU	700	0.0006
MTBench	1056	0.0009
MTBench_X	2	0.0
MultiBlimp	203	0.0002
Multi_IFEval	0	0.0
OpenBookQA	6	0.0
PAWS_x	12	0.0
PIQA	4	0.0
PolyMath	10	0.0
ToxiGen	10	0.0
XNLI	9	0.0
XStoryCloze	20	0.0
XWinograd	38	0.0
XWinogradNoBlanks	4	0.0

Table 10: Per benchmark contamination summary for the pes2o subset of OLMo-mix dataset for English. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	0	0.0
AIME_I_25	0	0.0
AlpacaEval	14	0.0
ArcChallenge	0	0.0
ArcEasy	22	0.0
Belebele	248	0.0002
BoolQ	6994	0.006
COPA	0	0.0

Benchmark	Contaminated Matches	Contamination %
CommonsenseQA	0	0.0
DoNotAnswer	0	0.0
Flores200	24	0.0
GPQA	0	0.0
GSM8K	0	0.0
HellaSwag	2	0.0
Hendrycks_MATH	42	0.0
HumanEval	18	0.0
HumanEval_Plus	0	0.0
IFEval	8	0.0
Lambada	0	0.0
m_ArenaHard	0	0.0
MBPP	0	0.0
MBPP_Plus	0	0.0
MMLU	360	0.0003
MTBench	234	0.0002
MTBench_X	0	0.0
MultiBlimp	26	0.0
Multi_IFEval	0	0.0
OpenBookQA	0	0.0
PAWS_x	2794	0.0024
PIQA	0	0.0
PolyMath	2	0.0
ToxiGen	0	0.0
XNLI	2	0.0
XStoryCloze	0	0.0
XWinograd	2	0.0
XWinogradNoBlanks	0	0.0

Table 11: Per benchmark contamination summary for the Wiki subset of OLMo-mix dataset for English. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	0	0.00000
AIME_I_25	0	0.00000
AlpacaEval	170	0.00000
ArcChallenge	109	0.00000
ArcEasy	157	0.00000
Belebele	513	0.00000
BoolQ	23275	0.00015
COPA	0	0.00000
CommonsenseQA	0	0.00000
DoNotAnswer	1	0.00000
Flores200	51	0.00000
GPQA	0	0.00000
GSM8K	11	0.00000
HellaSwag	3034	0.00002
Hendrycks_MATH	1065	0.00001
HumanEval	20	0.00000
HumanEval_Plus	0	0.00000
IFEval	59	0.00000
Lambada	28	0.00000
MBPP	27	0.00000
MBPP_Plus	5	0.00000
MMLU	2841	0.00002
MTBench	3127	0.00002
MTBench_X	44	0.00000
MultiBlimp	135	0.00000
Multi_IFEval	2	0.00000
OpenBookQA	1	0.00000

Benchmark	Contaminated Matches	Contamination %
PAWS_x	136	0.00000
PIQA	2	0.00000
PolyMath	64	0.00000
ToxiGen	16	0.00000
XNLI	31	0.00000
XStoryCloze	0	0.00000
XWinograd	31	0.00000
XWinogradNoBlanks	0	0.00000
m_ArenaHard	14	0.00000

Table 12: Per benchmark contamination summary for MultiSynt dataset. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	0	0.0000
AIME_I_25	0	0.0000
AlpacaEval	70	0.0010
ArcChallenge	113	0.0017
ArcEasy	139	0.0021
Belebele	8	0.0001
BoolQ	5142	0.0768
COPA	0	0.0000
CommonsenseQA	2	0.0000
DoNotAnswer	0	0.0000
Flores200	0	0.0000
GPQA	0	0.0000
GSM8K	129	0.0019
HellaSwag	119	0.0018
Hendrycks_MATH	7830	0.1169
HumanEval	381	0.0057
HumanEval_Plus	4	0.0001
IFEval	6	0.0001
Lambada	0	0.0000
MBPP	177	0.0026
MBPP_Plus	33	0.0005
MMLU	718	0.0107
MTBench	2746	0.0410
MTBench_X	136	0.0020
MultiBlimp	3	0.0000
Multi_IFEval	0	0.0000
OpenBookQA	0	0.0000
PAWS_x	48	0.0007
PIQA	16	0.0002
PolyMath	809	0.0121
ToxiGen	1	0.0000
XNLI	3	0.0000
XStoryCloze	0	0.0000
XWinograd	1	0.0000
XWinogradNoBlanks	0	0.0000
m_ArenaHard	69	0.0010

Table 13: Per benchmark contamination summary for Finemath dataset. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	0	0.0000

Benchmark	Contaminated Matches	Contamination %
AIME_I_25	0	0.0000
AlpacaEval	5	0.0000
ArcChallenge	23	0.0000
ArcEasy	147	0.0003
Belebele	0	0.0000
BoolQ	207	0.0004
COPA	0	0.0000
CommonsenseQA	0	0.0000
DoNotAnswer	0	0.0000
Flores200	0	0.0000
GPQA	0	0.0000
GSM8K	18	0.0000
HellaSwag	7	0.0000
Hendrycks_MATH	3737	0.0071
HumanEval	180	0.0003
HumanEval_Plus	1936	0.0037
IFEval	29	0.0001
Lambada	0	0.0000
MBPP	927	0.0018
MBPP_Plus	330	0.0006
MMLU	629	0.0012
MTBench	1028	0.0020
MTBench_X	613	0.0012
MultiBlimp	0	0.0000
Multi_IFEval	0	0.0000
OpenBookQA	0	0.0000
PAWS_x	0	0.0000
PIQA	0	0.0000
PolyMath	88	0.0002
ToxiGen	0	0.0000
XNLI	8	0.0000
XStoryCloze	0	0.0000
XWinograd	3	0.0000
XWinogradNoBlanks	0	0.0000
m_ArenaHard	12	0.0000

Table 14: Per benchmark contamination summary for Megamath dataset's text-code-block subset. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	3	0.0000
AIME_I_25	0	0.0000
AlpacaEval	130	0.0009
ArcChallenge	30	0.0002
ArcEasy	178	0.0012
Belebele	8	0.0001
BoolQ	3069	0.0205
COPA	0	0.0000
CommonsenseQA	1	0.0000
DoNotAnswer	0	0.0000
Flores200	11	0.0001
GPQA	2	0.0000
GSM8K	173	0.0012
HellaSwag	47	0.0003
Hendrycks_MATH	6995	0.0467
HumanEval	222	0.0015
HumanEval_Plus	20	0.0001
IFEval	11	0.0001
Lambada	0	0.0000
MBPP	313	0.0021

Benchmark	Contaminated Matches	Contamination %
MBPP_Plus	38	0.0003
MMLU	1531	0.0102
MTBench	2897	0.0193
MTBench_X	58	0.0004
MultiBlimp	1	0.0000
Multi_IFEval	0	0.0000
OpenBookQA	0	0.0000
PAWS_x	30	0.0002
PIQA	0	0.0000
PolyMath	1320	0.0088
ToxiGen	1	0.0000
XNLI	3	0.0000
XStoryCloze	0	0.0000
XWinograd	0	0.0000
XWinogradNoBlanks	0	0.0000
m_ArenaHard	130	0.0009

Table 15: Per benchmark contamination summary for Megamath dataset's web-pro subset. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

Benchmark	Contaminated Matches	Contamination %
AIME_II_25	0	0.0000
AIME_I_25	0	0.0000
AlpacaEval	241	0.0001
ArcChallenge	6	0.0000
ArcEasy	9	0.0000
Belebele	12	0.0000
BoolQ	1455	0.0007
COPA	1	0.0000
CommonsenseQA	7	0.0000
DoNotAnswer	1	0.0000
Flores200	2	0.0000
GPQA	7	0.0000
GSM8K	0	0.0000
HellaSwag	17	0.0000
Hendrycks_MATH	332	0.0002
HumanEval	223	0.0001
HumanEval_Plus	167	0.0001
IFEval	5	0.0000
Lambada	304	0.0001
MBPP	496	0.0002
MBPP_Plus	56	0.0000
MMLU	988	0.0005
MTBench	575	0.0003
MTBench_X	111	0.0001
MultiBlimp	91	0.0000
Multi_IFEval	0	0.0000
OpenBookQA	2	0.0000
PAWS_x	11	0.0000
PIQA	1	0.0000
PolyMath	15	0.0000
ToxiGen	1	0.0000
XNLI	16	0.0000
XStoryCloze	0	0.0000
XWinograd	29	0.0000
XWinogradNoBlanks	9	0.0000
m_ArenaHard	579	0.0003

Benchmark	Contaminated Matches	Contamination %

Table 16: Per benchmark contamination summary for Starcoder dataset. Results shown are total amount of contaminated documents where at least one benchmark n-gram did not pass the set threshold, and the contamination ratio of each benchmark over the dataset.

A.2 Additional model evaluation results

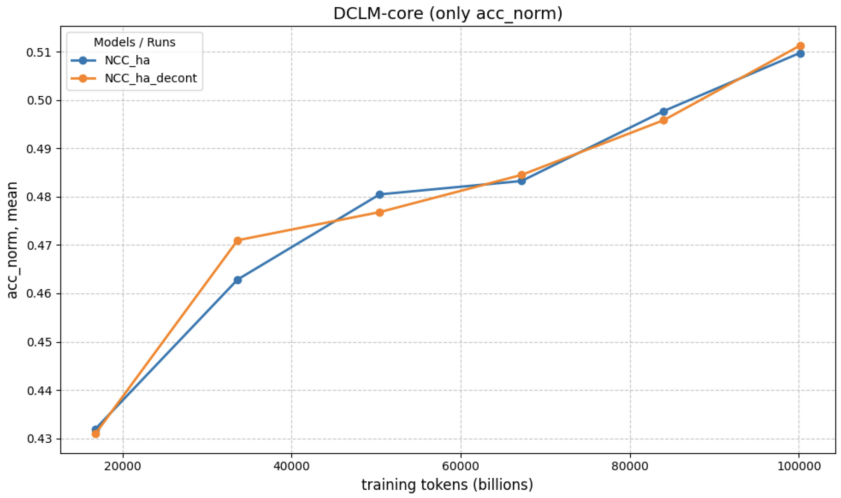


Figure 3: Aggregated performance comparison on Nemotron-CC with and without decontamination on the following dclm-core-22 evaluatin tasks: hellaswag (10 shot), hellaswag (0-shot), arc_easy, openbookqa, arc_challenge, piqa.

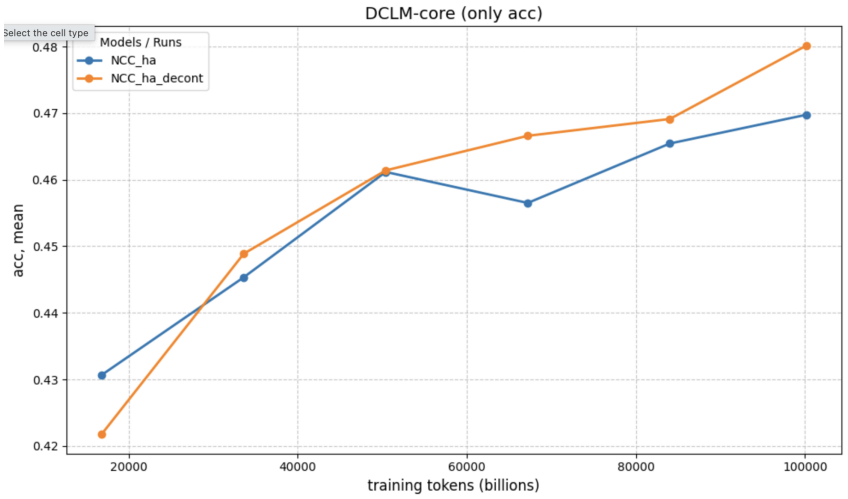


Figure 4: Aggregated performance comparison on Nemotron-CC with and without decontamination on the following dclm-core-22 evaluatin tasks: copa, commonsense_qa, agieval_lsat_ar ,wsc273, boolq, lambada_openai, winogrande.

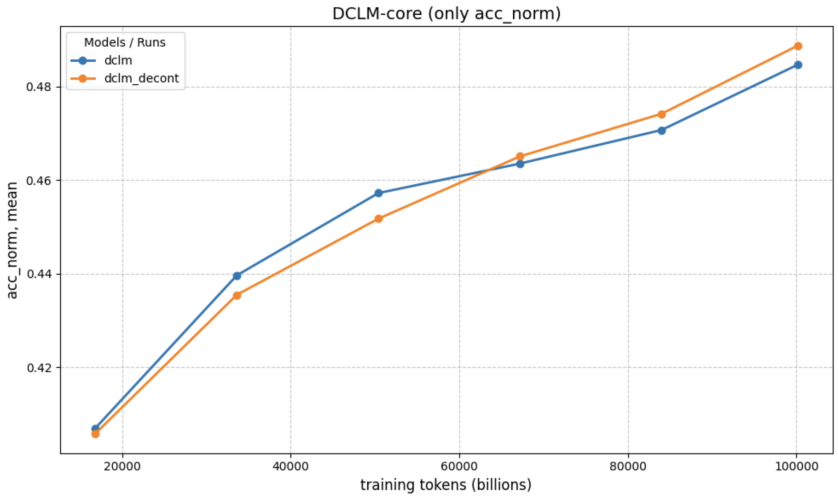


Figure 5: Aggregated performance comparison on DCLM with and without decontamination on the following dclm-core-22 evaluatin tasks: hellaswag (10 shot), hellaswag (0-shot), arc_easy, openbookqa, arc_challenge, piqa.

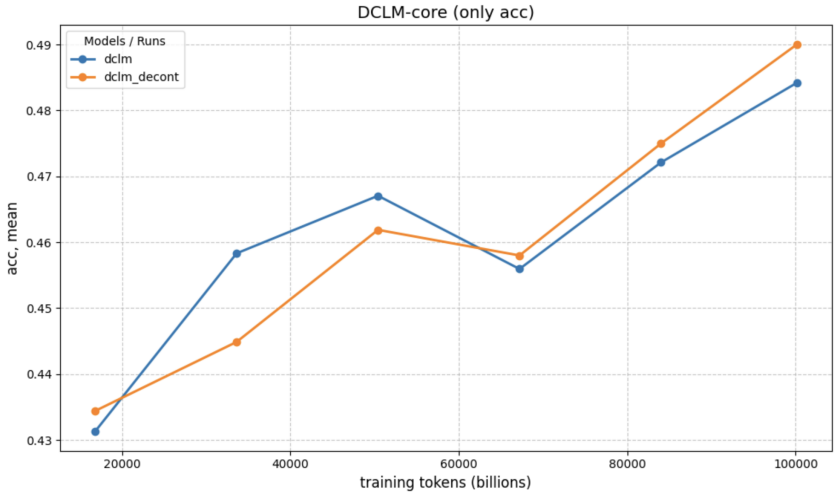


Figure 6: Aggregated performance comparison on DCLM with and without decontamination on the following dclm-core-22 evaluatin tasks: copa, commonsense_qa, agieval_lsat_ar ,wsc273, boolq, lambada_openai, winogrande.

dataset_name	passages	matches	matches_prc	unique_matches	unique_matches_prc
AIME_25	30	0	0	0	0.00
AlpacaEval	1593	170	0	44	2.76
ArcChallenge	3516	109	0	81	2.30
ArcEasy	7128	157	0	120	1.68
Belebele	29700	513	0	245	0.82
BoolQ	3270	23275	1.5e-04	6533	199.79
COPA	1100	0	0	0	0.00
CommonsenseQA	1221	0	0	0	0.00
DoNotAnswer	939	1	0	1	0.11
Flores200	37444	51	0	25	0.07
GPQA	546	0	0	0	0.00
GSM8K	1319	11	0	9	0.68
HellaSwag	10042	3034	2.0e-05	1981	19.73
Hendrycks_MATH	5003	1065	1.0e-05	632	12.63
HumanEval	328	20	0	6	1.83
IFEval	541	59	0	22	4.07
Lambada	5153	28	0	24	0.47
MBPP	1352	32	0	19	1.41
MMLU	14042	2841	2.0e-05	1239	8.82
MTBench	1388	3127	2.0e-05	523	37.68
MTBench_X	400	44	0	5	1.25
MultiBlimp	60693	135	0	51	0.08
Multi_IFEval	22505	2	0	1	0.00
OpenBookQA	2000	1	0	1	0.05
PAWS_x	14000	136	0	78	0.56
PIQA	5512	2	0	2	0.04
PolyMath	3000	64	0	37	1.23
ToxiGen	940	16	0	13	1.38
XNLI	27449	31	0	31	0.11
XStoryCloze	7555	0	0	0	0.00
XWinograd	2671	31	0	7	0.26
m_ArenaHard	6000	14	0	9	0.15

Table 17: AUXAUX